Ex-8
1/10/2026

NMAP to Discover LIVE HOST USING NMAP SCANS, (ARP, ICMP, TCP/UDP) ON THE TRYHACKME PLATFORM

AIM:-

To discover live hosts on a network using Nmap host-discovery scans - ARP SCAN, ICMP, TCMP, TCP/PUDPR ping scan - and to understand which protocol operates at which layer of TCP/LCP model and cut on each one applies.

INTRODUCTION:-

→ Nmap is a free open-source network scanning tool

→ It is the industry standard tool for mapping networks, discovering live hosts

⇒ This lab focuses on list of two questions asked during reconnaissance:

1) Which systems are up? - answered in this room
2) What services are running on those systems? - answered in the follow-on Nmap port-scanning rooms

⇒ Npm can determine whether a host is alive using four different protocols

* ARP (Link layer) * TCPC (Transport layer)

* ICMP (Network layer) * UDP (Transport layer)

ARP Host Discovery

* On a local Ethernet, communication requires the destination MAC address.
* To learn it, a host broadcasts an ARP request any host holding that IP replies with MAC address.
* A reply proves the host is up.
* ARP only works within your own screenshot.

ICMP Host Discovery:-

* Request several request/replay pairs that can reveal a live host.
* Windows system & many firewalls block ICMP.

7.1 ICMP Echo (-PE)

```
$ Sudo nmap -PE -sn 10.200.6.0/24
Nmap scan report for 10.200.6.1
Host is up (0.069s latency)
Nmap scan report for 10.200.6.50
Host is up (0.070s latency)
Nmap scan report for 10.200.6.250
Host is up (0.069s latency)
Nmap done: 256 IP addresses (3 hosts up) scanned in 3.38 secon
```

7.2 ICMP Timestamp (-PP)

```
$ nmap -PP -sn 10.200.6.0/24
Nmap scan report for 10.200.6.50
Host is up (0.069s latency).
Nmap scan report for 10.200.6.250.
Host is up (0.069s latency)
Nmap done: 256 IP addresses (2 hosts up) scanned in
8.44 seconds.
```

7.3 ICMP Address Mask (-PM)

```
$ nmap -PM -sn 10.200.6.0/24
Nmap done: 256 IP addresses (0 hosts up) scanned in 52.14 seconds
```

TCP and UDP Ping Discovery :-

* When ICMP is filtered, Nmap can send crafted TCP or UDP packets and infer that a host is alive purely from the fact that it responds at all - regardless of whether the probed ~~problem~~ port is open / closed.

8.1 TCP SYN PIN (-PS)

* Send a TCP packet with the SYN flag to target port. PN open port answers SYN|ACK; a closed port answers RST - either reply confirms the host is UP

```
$ nmap -PS -sn 10.200.6.0/24
Nmap scan report for 10.200.6.50
Host is up (0.070s latency).
Nmap scan report for 10.200.6.250
Host is up (0.068 s latency)
Nmap done: 256 IP addresses (2 hosts up) scanned in 33.88 seconds
```

8.2 TCP ACK ping (-PA)

```
$sudo nmap -PA -sn 10.200.6.0/24
Nmap scan report for 10.200.6.50
Host is up (0.070s latency).
Nmap scan report for 10.200.6.250
Host is up (0.069s latency).
Nmap done: 256 IP addresses (2 hosts up) scanned in
        36.64 seconds.
```

RESULT:-

* Nmap discover live host using nmap has been done successfully.